Arjuna JEE 2025

Maths

Permutations and Combinations

DPP: 3

Q1 Find the number of ways in which 5 boys and 5 girls be seated in a row, so that
(i) no two girls may sit together
(ii) all the girls sit together and all the boys sit together
(iii) all the girls are never together

Q2 Find the total number of 9 digit numbers which have all the digits different
(A) $9 \times 9!$ (B) $9!$
(C) $10!$ (D) None of these

Q3 A servant has to post 5 letters and there are 4 letter boxes. In how many ways can he post the letters is
(A) ${}^{5}P_{4}$
(B) $\frac{5!}{4!}$
(C) 5^{4}
(D) 4^{5}

Q4 In how many ways 5 delegates can be put in 6 hotels of a city if there is no restriction is
(A) 6^{5}
(B) 5^{6}
(C) ${}^{6}P_{5}$
(D) $\frac{6!}{5!}$

Q5 In a dinner party there are 10 Indians, 5 Americans and 5 Englishmen. In how many ways can they be arranged in a row so that all persons of the same nationality sit together
(A) $10!\ 5!\ 5!$
(B) $3!\ 10!\ 5!\ 5!$
(C) $3!\ 10!$
(D) $10!\ 5!$

Q6 In a class of students there are 4 girls and 6 boys. In how many ways can they be seated in a row so that all the four girls are not together
(A) $10! - 7!\ 4!$
(B) ${}^{10}P_{4}$
(C) ${}^{7}P_{4}$
(D) $6!\ 4!$

Q7 The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is
(A) 576 (B) 140
(C) 356 (D) 209

Q8 It is required to seat 5 men and 4 women in a row, so that the women occupy the even places, the number of such arrangements are
(A) 2880
(B) $5! \times 4$
(C) $5 \times 4!$
(D) none of these

Q9 The number of ways in which the letters of the word 'MIRACLE' can be arranged, if vowels always occupy the odd places, is
(A) 144 (B) 576
(C) $7!$ (D) 720

Q10 The number of permutations of the letters of the word CONSEQUENCE in which all the three E's are together, is

(A) $9!3!$

(B) $\frac{9!}{2!2!}$

(C) $\frac{9!}{2!2!3!}$

(D) $\frac{9!}{2!3!}$

Q11 How many words can be made with letters of the word INTERMEDIATE if no vowel is between two consonants

(A) 151200 (B) 164000

(C) 144200 (D) none of these

Q12 The number of ways in which six '$+$' and four '$-$' signs can be arranged in a line such that no two '$-$' signs occur together

(A) 24 (B) 35

(C) 30 (D) none

Q13 How many different words can be formed by using all the letters of the word 'ALLAHABAD' such that vowels occupy the even positions, is

(A) 30 (B) 40

(C) 50 (D) 60

Q14 If the four-letter words (need not be meaningful) are to be formed using the letters from the word "MEDITERRANEAN" such that the first letter is R and the fourth letter is E, then the total number of such words, is

(A) $\frac{11!}{(2!)^3}$

(B) 59

(C) 110

(D) 56

Q15 How many words are formed if the letters of the word GARDEN are arranged with the vowels in alphabetical order?

(A) 120 (B) 240

(C) 360 (D) 480

Q16 The number of arrangements of the letters of the word 'Delhi' if e always comes before i is

(A) 40 (B) 30

(C) 60 (D) 20

Q17 The number of arrangements of the letters of the word BANANA in which two N's do not appear adjacently is

(A) 40 (B) 60

(C) 80 (D) 100

Answer Key

Q1 (i) $5! \times 6!$ ways
(ii) $2!\,(5!)^2$ ways
(iii) $10! - 6!5!$ way

Q2 (A)

Q3 (D)

Q4 (A)

Q5 (B)

Q6 (A)

Q7 (A)

Q8 (A)

Q9 (B)

Q10 (B)

Q11 (A)

Q12 (B)

Q13 (D)

Q14 (B)

Q15 (C)

Q16 (C)

Q17 (A)